IGCSE 0607 Practice Questions — E3.6

Transforming graphs of functions

Total Questions: 12 — **Total Marks:** 75 Marks

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1 A function is defined as $y = f(x)$.

(a) Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = f(x - 4)$.

.....

..... **[2]**

(b) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ 7 \end{pmatrix}$ to give the graph of $y = g(x)$.
Write down the expression for $g(x)$ in terms of $f(x)$.

$g(x) =$ ..... **[1]**

2 **(a)** The graph of $y = x^2$ is translated by the vector $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$ to map onto the graph of $y = g(x)$. Write down $g(x)$ in terms of x.

$g(x) =$ ..... **[1]**

(b) The graph of $y = x^2$ is translated by the vector $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$ to map onto the graph of $y = h(x)$. Write down $h(x)$ in terms of x.

$h(x) =$ ..... **[1]**

(c) Describe fully the single transformation that maps the graph of $y = x^2$ onto the graph of $y = (x + 4)^2 - 1$.

.....

.....

..... **[2]**

3 A function is defined as $f(x) = 2x - 5$.

(a) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ 7 \end{pmatrix}$ to give the graph of $y = g(x)$.
Find $g(x)$ in terms of x, giving your answer in its simplest form.

$g(x) =$ ..... **[2]**

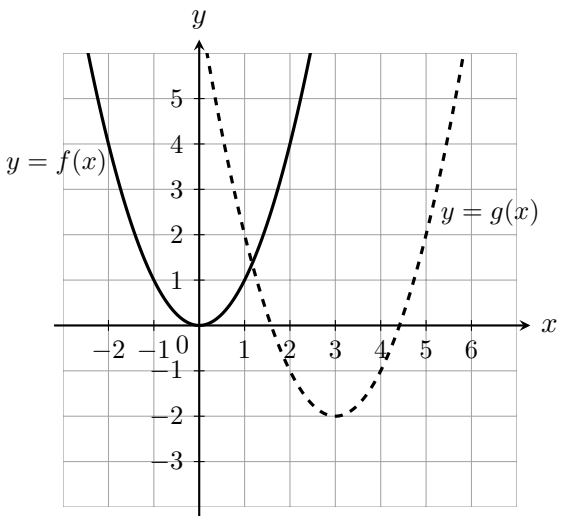
(b) Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = f(x - 4)$.

.....

.....

..... **[2]**

4 The diagram shows the graphs of $y = f(x)$ and $y = g(x)$ on a coordinate grid.



(a) Write down the column vector that describes the translation mapping the graph of $y = f(x)$ onto the graph of $y = g(x)$.

$$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$$ **[2]**

(b) Given that $f(x) = x^2$, write down $g(x)$ in the form $(x - a)^2 + b$.

$g(x) =$ ..... **[2]**

(c) Write down the coordinates of the vertex of the graph of $y = f(x + 5) - 1$.

(..... , ...) **[2]**

5 The graph of $y = f(x)$ has a local maximum at the point $P(-2, 5)$ and a local minimum at the point $Q(3, -1)$.

(a) Write down the coordinates of the local maximum of the graph of $y = f(x) - 4$.

(..... , ...) **[1]**

(b) Write down the coordinates of the local minimum of the graph of $y = f(x - 2)$.

(..... , ...) **[1]**

(c) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ such that the image of the local maximum P is at the origin $(0, 0)$.

(i) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

(ii) Write down the coordinates of the image of the local minimum Q under the same translation.

(..... , ...) **[1]**

6 Let $f(x) = 3x^2$.

(a) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$ to give the graph of $y = g(x)$.

Find $g(x)$ in terms of x.
Give your answer in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[3]**

(b) The graph of $y = g(x)$ is translated by the vector $\begin{pmatrix} 0 \\ -8 \end{pmatrix}$ to give the graph of $y = h(x)$.

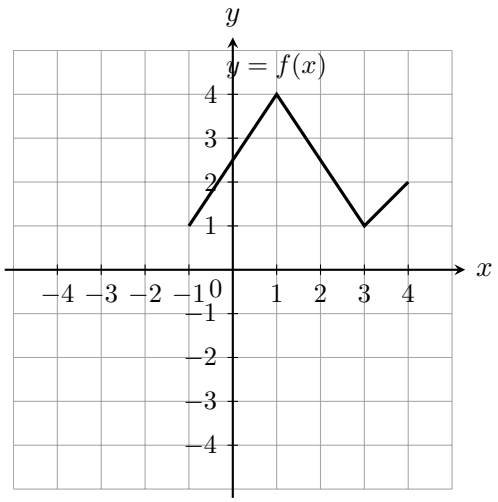
Write down the coordinates of the minimum point of the graph of $y = h(x)$.

(..... , ...) **[2]**

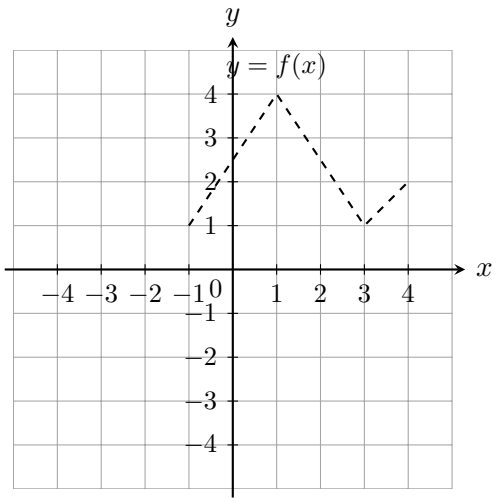
(c) Find the y-intercept of the graph of $y = h(x)$.

..... **[1]**

7 The diagram shows the graph of $y = f(x)$ on a coordinate grid.



(a) On the grid below, draw the graph of $y = f(x + 2) - 3$.



[2]

(b) The graph of $y = f(x)$ is mapped onto the graph of $y = h(x)$ by a single transformation.
The point $(1, 4)$ on $y = f(x)$ is mapped onto $(1, -1)$ on $y = h(x)$, and the point $(3, 1)$ is mapped onto $(3, -4)$.
Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = h(x)$.

.....

.....

..... **[2]**

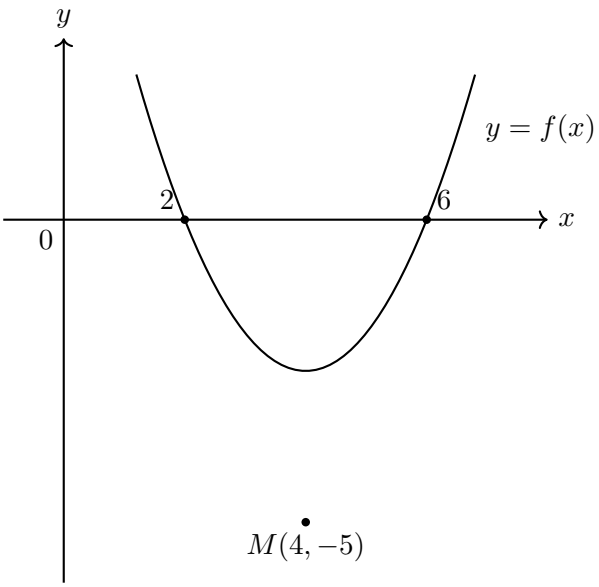
(c) Write down $h(x)$ in terms of $f(x)$.

$h(x) =$ ..... **[1]**

(d) Find the coordinates of the image of the point $(4, 2)$ on $y = f(x)$ when it is mapped onto the graph of $y = f(x - 3) + 4$.

(..... , ...) **[2]**

8 The diagram shows a sketch of the graph of $y = f(x)$.
The curve intersects the x-axis at $(2, 0)$ and $(6, 0)$ and has a minimum point at $M(4, -5)$.



(a) A transformation maps the graph of $y = f(x)$ onto the graph of $y = f(x) + 4$.
Write down the coordinates of the image of point M.

((..... , ...)) **[1]**

(b) The graph of $y = f(x)$ is mapped onto the graph of $y = g(x)$ by a translation with vector $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$.
Write down the expression for $g(x)$ in terms of $f(x)$.

$g(x) =$ ..... **[2]**

(c) The graph of $y = f(x - k)$ passes through the origin $(0, 0)$.
Find the two possible values of the constant k.

$k =$ ..... , $k =$ ..... **[2]**

(d) The function $h(x)$ is such that the graph of $y = h(x)$ has a minimum point at $(7, -5)$.
Find $h(x)$ in terms of $f(x)$.

$h(x) =$ ..... **[1]**

9 A function is given as $f(x) = x^2 - 2x - 3$.

(a) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -4 \\ 0 \end{pmatrix}$ to form the graph of $y = g(x)$.
Find $g(x)$ in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[4]**

(b) The graph of $y = f(x)$ is mapped onto the graph of $y = h(x)$ by a translation with vector $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$.
Solve the equation $h(x) = 0$.

$x =$ ..... , $x =$ ..... **[3]**

10 The functions $f(x)$ and $g(x)$ are defined as $f(x) = x^2 - 6x + 14$ and $g(x) = \dfrac{3}{x+2} - 5$.

(a) By writing $f(x)$ in the form $(x-a)^2 + b$, find the column vector of the translation that maps the graph of $y = x^2$ onto the graph of $y = f(x)$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[3]**

(b) The graph of $y = g(x)$ is translated by the vector $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ to map onto the graph of $y = h(x)$.
Find $h(x)$ in its simplest form.

$h(x) =$ ..... **[3]**

(c) Write down the equations of the vertical and horizontal asymptotes of the graph of $y = h(x)$.

$x =$ ..... , $y =$ ..... **[2]**

11 A function is defined as $f(x) = \dfrac{20}{x}$.

(a) The graph of $y = f(x)$ is mapped onto the graph of $y = g(x)$ by a translation with vector $\begin{pmatrix} 0 \\ -4 \end{pmatrix}$.

(i) Write down an expression for $g(x)$.

$g(x) =$ ..... **[1]**

(ii) Solve the equation $g(x) = 0$.

$x =$ ..... **[2]**

(b) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$ to give the graph of $y = h(x)$.
The graph of $y = h(x)$ passes through the point $(8, 5)$.
Work out the value of k.

$k =$ ..... **[3]**

(c) Let $p(x) = f(x - 3) + 7$.
Write down the equations of the vertical and horizontal asymptotes of the graph of $y = p(x)$.

Vertical asymptote: ..... , Horizontal asymptote: ..... **[2]**

12 The function f is defined by $f(x) = 2x^2 - 5x + 1$.

(a) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -2 \\ -6 \end{pmatrix}$ to give the graph of $y = g(x)$.

(i) Write down $g(x)$ in terms of $f(x)$.

$g(x) =$ ..... **[1]**

(ii) Find $g(x)$ in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[3]**

(b) The graph of $y = f(x)$ is mapped onto the graph of $y = h(x)$ by a single translation, where $h(x) = 2x^2 - 13x + 18$.
Find the column vector of this translation.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[4]**

(c) Find the coordinates of the point of intersection of the graphs $y = f(x)$ and $y = g(x)$.

(..... , ...) **[3]**